Juan David Muñoz-Bolaños
Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

Leica Microsystems / ZEISS
Hiring Team
Please update with actual address
City, Country

Subject: Application for System Developer Confocal Microscopy (d/f/m)

Dear Hiring Team,

I am writing to express my strong enthusiasm for the System Developer Confocal Microscopy position. As an Optical Engineer currently concluding my PhD in Innsbruck, my deep hands-on expertise in designing, assembling, and characterizing advanced nonlinear microscopy platforms directly aligns with your need for innovation in core confocal microscopy technologies.

In my current role, I take sole responsibility for the architecture and operational readiness of complex Two-Photon microscopy setups. I routinely integrate critical components—including high-power pulsed lasers, acousto-optic/spatial light modulators, and highly sensitive detection modules (PMTs/spectrometers)—to correct severe aberrations in deep biological tissue. This perfectly mirrors the hardware integration challenges required for next-generation point-scanning microscope development.

Advanced Microscopy & Technology Development:
During my PhD and my prior work at IPHT Jena, I have specialized in building optical prototypes from the ground up. I am highly comfortable evaluating new core concepts in the lab, from optimizing laser coupling and scanning algorithms to pushing the physical limits of signal-to-noise ratios in photon-starved imaging regimes.

Interdisciplinary Interface & System Architecture:
Beyond the optical bench, I understand that modern microscopy requires a seamless bridge between physics and electronics. I develop high-performance Python pipelines and hardware-control software to tightly synchronize laser excitation with real-time detection, ensuring robust and reliable system performance analogous to industrial R&D requirements.

Project Leadership & Problem Solving:
Through my work across academic research and industrial machine vision (INTECOL), I have developed a highly structured, analytical approach to problem-solving. I am accustomed to managing complex project deliverables independently while thriving in collaborative, interdisciplinary teams that drive economic and competitive technological advantages.

While my working language is English, I also hold a strong working proficiency in German (B2), enabling smooth communication across international and local teams. I would welcome the opportunity to discuss how my hands-on experience in multiphoton optical design and prototyping can directly contribute to your advanced microscopy development.

Sincerely,

Juan David Muñoz-Bolaños
Optical System Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer | Adaptive Optics

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145
juan.munoz@i-med.ac.at | LinkedIn Profile

PROFESSIONAL SUMMARY

Optical Engineer specialized in the architecture, integration, and prototyping of advanced nonlinear microscopy platforms. Extensive hands-on experience in Two-Photon microscopy, high-power laser integration, and precise metrology evaluation. Proven capability to bridge optical physics with hardware control software to drive R&D from conceptualization to reliable laboratory prototypes.

TECHNICAL COMPETENCIES

Advanced Microscopy & Optical Architecture
Substantial expertise in Two-Photon, confocal, and custom nonlinear microscopy principles. Demonstrated ability to design, assemble, and optimize complex optical platforms that push the physical limits of diffraction and deep tissue scaling.

Prototyping & Core Technologies
Hands-on integration of high-power continuous-wave/pulsed lasers, spatial light modulators, acousto-optics, and highly sensitive detection modules (PMTs, Spectrometers). Focus on signal-to-noise optimization.

Software Integration & Control
Advanced Python (JAX, NumPy) for hardware interfacing, data acquisition, and signal processing. Experience synchronizing scanning technologies with real-time detection software pipelines.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Engineering
Jan 2022 – Present

- **Microscopy System Architecture:** Designed and constructed advanced Two-Photon microscopy setups from the ground up, directing the full optical integration from laser source to detector.
- **Core Technology Integration:** Integrated and characterized novel core technologies, including adaptive optics hardware, to fundamentally improve spatial resolution in deep, scattering media.
- **Project Management:** Led independent R&D initiatives, taking sole operational responsibility for evaluating new concepts and maintaining complex laboratory prototypes.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Assistant Researcher
Jun 2020 – Dec 2021

- **Prototype Construction:** Assembled and calibrated vibrational point-scanning imaging prototypes, integrating fiber probes alongside sensitive detection setups.
- **Innovation & Evaluation:** Bridged theoretical R&D with practical lab implementation, utilizing structured data evaluation to streamline experimental characterization.

INTECOL S.A.S. — Medellin, Colombia
Machine Vision System Engineer
Mar 2018 – Mar 2019

- **System Reliability:** Ensured strict industrial robustness for optical inspection setups, gaining crucial experience in lifecycle management outside of academic settings.
- **Interdisciplinary Teamwork:** Collaborated heavily with external suppliers and cross-functional teams to integrate complex machine vision tracking systems.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. in Adaptive Optics & Microscopy *Jan 2022 – Present*

Friedrich Schiller University Jena Germany
M.Sc. in Photonics *2019 – 2021*

University of Cauca Colombia
B.Sc. in Physics Engineering *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. Applied Sciences, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte Medical University of Innsbruck
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich - Schiller - Universität Jena
2

Prof. Dr. Christian Spielmann
Dekan

Prof. Dr. Silvana Botti
Vorsitzende des Prüfungsausschusses

REPUBLICA DE COLOMBIA

PASAPORTE
PASSPORT

Tipo / *Type*: P
Cod. país / *Country code*: COL
Pasaporte Nº / *Passport No.*: AV410884

Apellidos / *Surname*
MUÑOZ BOLAÑOS

Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*: 04 JUN/JUN 1994
Núm. personal / *Personal No.*: CC1061772278

Sexo / *Sex*: M
Lugar de nacimiento / *Place of birth*: LA CRUZ COL

Fecha de expedición / *Date of issue*: 11 DIC/DEC 2018
Autoridad / *Authority*: G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*: 10 DIC/DEC 2028
Firma del titular / *Holder's signature*: Juan David M

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SONSTIGE ANGABEN
GILT BEI MEDIZINISCHE
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Juan David
GESCHLECHT
STAATSANGEHÖRIGKEIT
GEBURTSDATUM
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04 06 1994
KARTE GÜLTIG BIS
16 05 2026
ART DES AUFENTHALTSTITELS
NIEDERLASSUNGSBEWILLIGUNG
SONSTIGE ANGABEN
FORSCHER
349461
RESIDENCE PERMIT

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich • ösd INNSBRUCK • schweiz • deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen 1996

ÖSD ist Mitglied von

ALTE

ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Dr. Thomas Marschner
Program Manager Funding Projects

Dr. Thomas Stammler
CTO & Head of Product Strategy

ZEISS

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
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PBX. (574) 316 3322 Cell: (57) 301 430 2532

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LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020

F. Sondermann

Prof. J. Popp	F. Sondermann	Juan David Munoz Bolanos
Chairman	Executive Member	Guest
of Leibniz-IPHT	of Leibniz-IPHT	

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Jena, 13 November 2020

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest